

Abhiram Dodda

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EDUCATION

Pennsylvania State University | MS in Computer Science and Engineering **GPA: 3.89/4**
Relevant Coursework: Advanced Algorithms, HPC, Computer Architecture, Database Systems Aug 2025 – May 2027
GRIET | Bachelor of Technology in Computer Science and Engineering **GPA: 3.9/4.0**
Honors: Three-time Merit Endowment Award Recipient Dec 2021 – Jun 2025

EXPERIENCE

Software Development Intern May 2026 – August 2026
Amazon Web Services (AWS) *East Palo Alto, CA, USA*

- Built a production-faithful **discrete-event simulator** in **Kotlin** for AWS Redshift's **Capacity Management Service**, replaying **6.5M+ production events** through re-implemented capacity sweepers, a **15-state host lifecycle**, and the fulfillment chain to predict the cost and availability impact of config changes before production rollout.
- Validated the simulator against production end-state at **4.7% MAPE** on **ODCR reservation sizing** over a 7-day us-east-1 window by reverse-engineering sweeper and DynamoDB query semantics from production source code.
- Fixed **10 production-divergent modeling bugs**, including an attribute-exclusion bug governing **90%** of serverless-burst fleet retention and cross-plane demand clobbering affecting **14%** of demand keys.
- Designed a **Python** ingestion pipeline parsing **28M+ log lines** into structured events across multi-format DynamoDB exports, and shipped the simulator as a **Brazil package** with 71 tests and a run-comparison dashboard.

Software Development Intern May 2025 – Jul 2025
Platform3Solutions *Remote*

- Architected a self-hosted **Retrieval-Augmented Generation (RAG)** pipeline by deploying **Llama 3** via an **Ollama** server, utilizing **Flask** to facilitate **semantic search** for **500+** concurrent users with a mean query latency of **<200ms**.
- Optimized **retrieval precision by 76%** by engineering a **hierarchical metadata filtering** layer within **ChromaDB**, integrating **fine-grained RBAC** to ensure **100% data isolation** across multi-tenant project environments.
- Engineered **scalable, containerized Flask REST APIs** orchestrated via **Docker**, managing an **automated ingestion pipeline** that processed **5,000+ documents per hour** with **asynchronous local embedding** generation.

Data Science Intern Mar 2025 – Jun 2025
CaptainFresh *Remote*

- Executed comprehensive **Exploratory Data Analysis (EDA)** on **100k+** multi-band spectral observations, using **automated statistical profiling** to identify significant correlations ($r > 0.82$) between specific spectral bands and pond health indicators.
- Trained and benchmarked **Random Forest** and **XGBoost** models to predict pond ecosystem conditions, reaching an **R-squared of 0.88** through hyperparameter tuning and 5-fold cross-validation.

Research Assistant Sep 2023 – Mar 2025
GRIET *Hyderabad, India*

- Designed **high-precision diagnostic systems**, including a **CNN** for dermatological classification (**95.27% accuracy**) and a plant pathology framework reaching **96.82% top-1 accuracy** across **38** classes.
- Built a **multi-model ensemble** for **Optical Coherence Tomography (OCT)** analysis, fusing **VGG16**, **InceptionV3**, and **U-Net** to reach **98.86% accuracy** and an **AUC-ROC of 0.99**.
- Developed a patented assistive system for the visually impaired, integrating a **YOLO-based detector (30+ FPS)** with **Tesseract OCR** for sub-second auditory obstacle feedback.
- Mitigated severe **class imbalance** in medical datasets via a hybrid pipeline of **synthetic augmentation** and strategic under-sampling, improving minority-class **F1 by 14%**.

Research Intern May 2024 – Dec 2024
IIT Hyderabad *Hyderabad, India*

- Quantified **semantic stability** in **Transformers** via a proposed **semi-intrinsic evaluation metric**, analyzing probability-distribution shifts (median $\Delta \approx 0.03$) across 10^4 **synonym substitutions** to measure embedding robustness.
- **Fine-tuned BERT-large** for domain-specific **sentiment analysis**, achieving an **80% F1-score** and cutting **training convergence time by 20%** through hyperparameter optimization.
- Optimized **Information Retrieval** for judicial cross-referencing using **BM25** and **cosine similarity**, achieving a **Mean Reciprocal Rank of 0.82** across heterogeneous legal datasets.

PROJECTS

Distributed Image Processing Platform | Go, S3/MinIO, Kubernetes, Ray, gVisor, SQLite [Link](#)

- Built a distributed image-processing platform in **Go** with **256 SHA-256 hash-prefix shards** on S3, a zero-allocation **consistent-hash ring**, a Kubernetes operator, and **gVisor-sandboxed** execution of untrusted algorithms, sustaining

1.17 GB/s aggregate scan throughput.

- Chaos-tested fault tolerance on **1.28M real ImageNet objects** by killing **3 of 10 workers** mid-job, where **phi-accrual failure detection** rebalanced 52 in-flight tasks and all **256 tasks completed with zero lost or duplicated results**.
- Guaranteed **exactly-once shard processing** under failure via content-addressed sharding and **idempotent result keys** that make a rebalanced task's re-run overwrite rather than duplicate, backed by a **write-ahead log + checkpointing** so in-flight jobs survive coordinator restarts.
- Designed **intra-shard work stealing** with bounded lease grants – a live run tiled a 6,000-image hot shard across two workers in **53 steals with no gap or double-processing** – and added **per-tenant quotas**, token-bucket rate limiting, and **JWT auth with RBAC**, verified by **117 tests across 13 packages** under **-race**.

Distributed ML Inference Engine | C++17, ONNX Runtime, CUDA, Multi-threading

[Link](#)

- Engineered a **C++17** distributed ML inference system on **ONNX Runtime** with **CUDA** acceleration and CPU fallback, routing requests by a hash of input data so each result is computed and **cached exactly once cluster-wide**, serving **1M requests at a 100% success rate** behind a circuit-breaker gateway.
- Designed **SLO-aware adaptive batching** that flushes when the oldest queued request nears a **20ms latency budget**, growing batches from 8 to 31 under load while holding **p50 flat at ~120ms**.
- Ran a **p50–p99.9 tail-latency load sweep** that traced a 130 req/sec throughput ceiling to an undersized HTTP thread pool capping in-flight requests below the batch size, and right-sized the pools to scale the cluster **3.8x to 500 req/sec** at a **p99 of 324ms with zero failures**.

tensorforge – ONNX Inference Compiler | C++17, CUDA, cuBLAS, NVRTC, Python

[Link](#)

- Built an **ONNX-to-CUDA inference compiler**: an SSA **graph IR**, an optimizing **pass pipeline** (constant folding, DCE, operator fusion, and **liveness-based memory planning** via interval coloring), and a C++/CUDA runtime that calls cuBLAS for GEMMs and **NVRTC-generated fused kernels** for glue.
- Cut ResNet50 kernel launches **227 to 144** via conv-epilogue fusion (folding **Conv+BN+ReLU** into one kernel) and peak arena memory **12.6x** through the planner, at **bit-for-bit parity** with ONNX Runtime (max abs error **4.8e-6**); shipped as a drop-in backend of the inference engine below.

Speech to Sign Translation | Whisper, T5-Transformer, PyTorch, OpenCV, NLP

[Link](#)

- Architected a multimodal pipeline integrating **Whisper-Large-v3** for speech-to-text and a **fine-tuned T5-Transformer** for text-to-gloss translation.
- Achieved **77% translation accuracy** and a **BLEU score of 0.42**, benchmarking against standard ASL datasets, with a sub-second (**1.1s**) average inference time per sequence via **INT8 quantization** and optimized tokenization.

PUBLICATIONS & PATENTS

Publications

- **A Novel Deep-Learning Based Classification of Skin Diseases**
Tabitha Indupalli, Dodda Abhiram*, Rayapuraju Srivatsav, Singamsetty Aashrith, Vudiga Vineeth*
5th IEEE Global Conference for Advancement in Technology (GCAT), 2025
- **A Novel Two-Stage Deep Learning Framework for Plant Disease Detection**
Dodda Abhiram, K. Anuradha, V. Srilakshmi, K. Adilakshmi*
5th IEEE Global Conference for Advancement in Technology (GCAT), 2025
- **Deep Learning and OCT Imaging: A Novel Ensemble Approach for Eye Disease Diagnosis**
Dodda Abhiram, R. Aruna Flarance, V. Srilakshmi, K. Anuradha*
4th International Conference on Cognitive & Intelligent Computing (ICCIC) – Springer
- **Ensemble Approach for Blood Vessel Segmentation in Retinal Images**
Dodda Abhiram, Y. Ekantha Sai, B. Siddarth Mahesh, K. Anuradha, B. Sankara Babu*
International Conference on Intelligent Systems & Sustainable Computing (ICISSC) – Springer

Patent

- **Smart Glasses for Visually Impaired Individuals**
Sreejyothsna Ankam, Manav, Abhiram Dodda, Dinesh Chandra, Nagireddy Padmakshaya
Patent No. 202541000874 A, Issued Jan 11, 2025
Wearable assistive device with object detection, obstacle recognition, OCR-based reading, currency detection, and voice-controlled smartphone integration.

ACHIEVEMENTS

- **Merit Endowment Award** (2021–2022, 2022–2023, and 2025), Gokaraju Rangaraju Institute of Engineering and Technology – three-time recipient.
- Ranked in the **Top 5%** nationally, IIT Madras NPTEL Data Science for Engineers.

SKILLS

Languages:	Go, C/C++, Python, Java, Kotlin, SQL, Shell
Systems:	Consistent Hashing, WAL/Crash Recovery, Fault Detection, Exactly-Once Semantics, Multi-threading, gRPC, CUDA
Cloud & Data:	AWS (EC2/S3/Lambda), Kubernetes, Docker, gVisor, Ray, Linux, Terraform, PostgreSQL, NoSQL (MongoDB, Redis), Spark
ML:	PyTorch, TensorFlow, JAX, ONNX, scikit-learn, pandas, NumPy